

BARABOO WISCONSIN DELLS, WI, USA

WMO: 726503

Lat: 43.522N

Lon: 89.774W

Elev: 976

StdP: 14.18

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54833

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.7	-2.4	-16.7	2.3	-5.8	-11.3	3.1	-0.8	21.1	23.2	19.5	23.3	4.0	280	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	88.9	73.8	85.9	71.7	82.8	70.4	77.3	84.2	75.2	82.0	73.3	80.1	10.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	136.6	81.0	72.9	126.5	79.2	71.7	121.3	78.0	41.5	83.2	39.3	81.7	37.5	79.6	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	18.0	16.2	-15.2	93.3	5.5	3.5	-19.2	95.8	-22.4	97.9	-25.5	99.8	-29.5	102.4		
			-15.5	79.5	5.2	3.3	-19.3	81.8	-22.3	83.7	-25.2	85.6	-29.0	87.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	19.4	22.7	34.1	46.4	57.8	67.1	71.4	69.3	62.3	49.6	36.9	24.5
	DBStd	20.53	12.12	11.88	11.85	9.71	8.60	6.66	5.72	5.41	8.12	9.31	10.08	11.41
	HDD50	3753	948	765	512	177	23	0	0	0	6	124	407	791
	HDD65	7187	1413	1184	958	560	257	53	11	20	145	487	843	1256
	CDD50	2632	0	1	21	70	265	513	664	597	375	110	15	1
	CDD65	590	0	0	1	4	34	116	211	152	64	8	0	0
	CDH74	5127	0	0	7	54	333	1034	1912	1180	548	59	0	0
	CDH80	1419	0	0	0	9	78	278	609	301	138	6	0	0
Wind	WSAvg	6.7	7.0	7.1	7.3	8.2	7.4	6.2	5.4	4.8	5.8	6.9	7.4	6.9
Precipitation	PrecAvg	36.3	1.3	1.2	2.1	3.8	4.2	5.6	4.4	4.5	3.3	2.4	2.0	1.6
	PrecMax	46.7	2.8	2.9	5.8	7.2	11.3	13.5	11.0	13.7	11.1	4.9	4.8	3.7
	PrecMin	26.1	0.4	0.0	0.4	0.5	1.7	0.6	1.4	1.4	0.4	0.8	0.3	0.2
	PrecStd	5.9	0.6	0.7	1.3	1.7	1.9	3.1	2.4	2.6	2.2	1.3	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	56.2	73.0	80.8	86.9	90.4	94.8	90.5	89.2	81.0	68.4	56.7
		MCWB	41.8	49.5	60.1	61.5	68.3	73.0	76.4	74.5	72.4	69.3	57.3	54.1
	2%	DB	42.9	48.0	63.9	73.0	81.6	86.3	89.8	86.4	83.9	74.6	62.6	48.0
		MCWB	37.9	42.5	54.1	58.7	66.1	71.0	76.0	72.6	70.1	64.2	54.4	44.3
	5%	DB	37.2	43.1	57.3	68.3	77.6	82.5	86.1	82.8	80.6	70.1	56.8	43.1
		MCWB	34.5	39.2	49.2	54.5	62.8	69.6	72.8	70.8	68.1	61.6	50.4	39.2
	10%	DB	34.9	38.1	51.8	63.4	73.1	80.6	82.7	80.9	76.6	64.5	52.3	38.7
		MCWB	33.3	34.9	44.6	51.5	61.2	68.4	71.5	69.5	65.3	56.8	46.3	35.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	49.6	62.5	65.3	73.3	77.6	83.5	78.6	75.3	72.9	60.0	54.7
		MCDB	46.3	54.5	71.5	75.9	82.4	85.3	88.9	86.2	84.1	78.9	66.2	56.4
	2%	WB	38.1	43.3	57.3	60.6	68.9	75.0	78.8	75.9	72.7	67.0	55.8	45.3
		MCDB	42.1	47.2	62.5	70.5	77.7	82.7	84.8	82.7	80.2	71.7	60.7	46.5
	5%	WB	35.4	39.3	50.4	57.0	66.3	72.8	75.9	73.8	70.3	62.1	51.2	39.6
		MCDB	37.4	43.3	57.1	65.7	74.8	80.5	82.3	80.0	77.2	67.2	55.9	42.8
	10%	WB	33.2	35.6	44.7	53.0	63.5	70.2	73.8	71.8	67.9	58.4	46.3	36.1
		MCDB	34.8	38.4	50.4	61.1	70.3	77.5	80.4	77.8	74.2	64.6	51.0	38.6
Mean Daily Temperature Range	5% DB	MDBR	14.3	15.5	17.2	19.7	20.3	19.7	20.1	19.4	20.9	18.1	14.8	13.4
		MCDBR	14.9	17.3	23.0	27.2	24.7	22.6	22.2	21.8	23.6	23.5	21.4	16.7
		MCWBR	12.2	12.9	14.3	15.2	12.6	11.2	10.5	10.4	11.7	14.1	14.9	13.3
	5% WB	MCDBR	13.7	16.5	20.7	23.5	20.9	19.7	18.9	18.4	20.2	19.8	19.7	15.3
		MCWBR	11.5	12.7	14.4	15.4	12.3	11.7	11.7	10.8	11.8	14.5	15.7	13.3
Clear-Sky Solar Irradiance	taub	0.270	0.311	0.327	0.361	0.390	0.405	0.405	0.398	0.363	0.326	0.291	0.269	
	taud	2.383	2.262	2.377	2.335	2.302	2.294	2.297	2.312	2.401	2.501	2.525	2.447	
	Ebn at Noon	276	278	290	286	280	275	273	272	274	272	268	266	
	Edh at Noon	24	33	34	38	41	41	40	38	33	26	21	21	
All-Sky Solar Radiation	RadAvg	545	884	1222	1429	1703	1891	1925	1660	1346	864	565	410	
	RadStd	67	83	145	129	128	138	95	112	109	103	50	38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page